Input: KS orbitals $\psi_n(\mathbf{r}) = \sum_\mu C_{\mu n}\,\phi_\mu(\mathbf{r})$, energies ε_n, and atom-centered grid points $\{\mathbf{r}_l\}$

1 Fitting coefficients (for every atom A) $O(N)$

$$\sum_{l'} D_{ll'}\,Z_{l'P} = d_{lP}\,, \quad D_{ll'} = \Big[\sum_\mu \phi_\mu(\mathbf{r}_l)\,\phi_\mu(\mathbf{r}_{l'})\Big]^2,$$

$$d_{lP} = \sum_{\mu\nu} \phi_\mu(\mathbf{r}_l)\,\phi_\nu(\mathbf{r}_l)\,(\mu\nu|P)$$

2 Polarizability (for every imaginary time τ) $O(N)$

$$\chi_{PQ}(i\tau) = -2\sum_{ll'} Z_{lP}\,G_{ll'}(i\tau)\,G_{l'l}(-i\tau)\,Z_{l'Q}\,,$$

$$G_{ll'}(i\tau) = \sum_{\mu\nu} \phi_\mu(\mathbf{r}_l)\,G_{\mu\nu}(i\tau)\,\phi_\nu(\mathbf{r}_{l'})$$

3 Screened interaction (for every frequency ω) $O(N^3)$

$$\boldsymbol{\epsilon}(i\omega) = \mathbf{1} - \mathbf{V}^{1/2}\mathbf{M}^{-1}\boldsymbol{\chi}(i\omega)\mathbf{M}^{-1}\mathbf{V}^{1/2},$$

$$\widetilde{\mathbf{W}}^{\mathrm{c}}(i\omega) = \mathbf{M}^{-1}\mathbf{V}^{1/2}\big[\boldsymbol{\epsilon}^{-1}(i\omega) - \mathbf{1}\big]\mathbf{V}^{1/2}\mathbf{M}^{-1}$$

4 Self-energy (for every imaginary time τ) $O(N)$

$$\Sigma^{\mathrm{c}}_{\mu\nu}(i\tau) = i\sum_{ll'} \phi_\mu(\mathbf{r}_l)\,G_{ll'}(i\tau)\,\widetilde{W}^{\mathrm{c}}_{ll'}(i\tau)\,\phi_\nu(\mathbf{r}_{l'})\,,$$

$$\widetilde{W}^{\mathrm{c}}_{ll'}(i\tau) = \sum_{PQ} Z_{lP}\,\widetilde{W}^{\mathrm{c}}_{PQ}(i\tau)\,Z_{l'Q}$$

5 Quasiparticle energies (analytic continuation) $O(N^3)$

$$\Sigma^{\mathrm{c}}_{nn}(i\tau) = \sum_{\mu\nu} C_{\mu n}\,\Sigma^{\mathrm{c}}_{\mu\nu}(i\tau)\,C_{\nu n}\,,$$

$$\varepsilon_n^{G_0W_0} = \varepsilon_n + \Sigma^{\mathrm{x}}_{nn} + \mathrm{Re}\,\Sigma^{\mathrm{c}}_{nn}\big(\varepsilon_n^{G_0W_0}\big) - v^{\mathrm{xc}}_{nn}$$

$O(N^2)$ per state n, $O(N^3)$ for all states

sparse contraction dense, auxiliary / MO basis

matrices in green are sparse